

Emma Bennett

571-474-5696 | emmabennett135@gmail.com | [linkedin.com/in/emma-bennett4](https://www.linkedin.com/in/emma-bennett4)

EDUCATION

Duke University

Durham, NC

B.S. in Computer Science and Electrical & Computer Engineering, Concentration in AI/Machine Learning May 2027

- Relevant coursework: Deep Learning, Applied Machine Learning, Computer Vision, Linear Algebra, Probability

WORK EXPERIENCE

LinkedIn

Sunnyvale, CA

Software Engineer Intern

May 2026 - August 2026

- Built an LLM-based autofixer using LangChain that detects sync-blocking calls in async Python code and auto-generates fixes. Reduced event-loop stalls by 85% (10K→1.3K/day) in a production backend service.
- Scaled fix correctness evaluation via a Kafka-to-Pinot telemetry pipeline, resolving hundreds of violations across 17 internal services at an 89.8% average fix score.

Software Engineer Intern

May 2025 - August 2025

- Designed and developed data lake to enable security monitoring of previously siloed 3rd-party applications within CrowdStrike's Adaptive Shield platform using Azure, Databricks, Apache Spark, and Python.
- Built an automated ticket workflow system in Python between Adaptive Shield and ServiceNow to open/close tickets based on security checks, eliminating manual ticketing overhead.

Duke University School of Medicine

Durham, NC

Undergraduate Research Assistant

September 2024 - Present

- Developing a reproducible deep learning benchmarking framework to evaluate pooling strategies across protein-language models (PLMs) for protein fitness prediction under Professor Navid NaderiAlizadeh.
- Implementing mean, max, top-k, light attention, SWE, and BOM pooling in a unified training pipeline, measured via Spearman correlation across various PLM architectures, seeds, and LoRA ranks.

MITRE Corporation

McLean, VA

Software Engineer Intern

May 2024 - August 2024

- Designed and implemented cyber agents that generate and execute action plans in simulation and live-fire environments by adapting AI planning algorithms for the cybersecurity domain. Used Python and Jupyter Notebook.

Software Engineer Intern

June 2023 - August 2023

- Researched and implemented a cyber agent architecture where communications among agents are entirely driven by LLMs on a GPU cluster. Agents communicate to search the network topology and capture a flag. Used Python in Jupyter Notebook and experimented with several LLMs. Results presented at Black Hat 2024.

Research Intern

June 2022 - August 2022

- Researched and implemented in Go a new adversarial cost algorithm based on A*, factoring in weighted cost from detected adversarial cyber agent nodes on networks. Also implemented a network simulation tool in NetworkX.

Pacific Northwest National Laboratory

Richland, WA (remote)

Science Programs Intern

June 2021 - August 2021

- Analyzed Zeek logs, a large-scale dataset from a network security monitoring tool, and performed anomaly detection utilizing several machine learning algorithms. Used Python and Scikit Learn.

TECHNICAL SKILLS

Languages: Python, Java, C, Go, SQL **AI/ML:** LLMs, LangChain, PyTorch, TensorFlow, Scikit-learn, NumPy **Infra/Data:** Kafka, Apache Pinot, Databricks, Apache Spark, Azure, Docker, OpenTelemetry **Platforms:** GitHub, Linux, Windows

EXTRACURRICULAR ACTIVITIES

Duke Motorsports FSAE Team (Electrics Subsystem)

September 2023 - Present

- Led the design and iteration of high-reliability PCBs for the vehicle steering interface, applying aerospace-grade design practices (signal integrity, thermal management).
- Integrated embedded control units (ECUs) with custom wiring harnesses and battery subsystems, delivering a low-latency, fault-tolerant control architecture.